Additional Table 1 Characteristics of the included studies

Study	Country	Study group	N	Sex (F/M)	Age (years)	MMSE (scores)	Biomarker	Sample source	Isolation	Analytical method	Study type
Cheng et al., 2015	Australia (AIBL)	AD	23;16	13/9;10/5	78.89±7.33; 78.29±7.55	15±8.66; 21±8.14	hsa-miR-361-5p, hsa-miR-30e-5p, hsa-miR-93-5p, hsa-miR-15a-5p, hsamiR-143-3p, hsa-miR-335-5p, hsa-miR-106b-5p, hsa-miR-101-3p, hsa-miR-424-5p, hsa-miR-106a-5p, hsa-miR-18b-5p, hsa-miR-3065-5p, hsa-miR-20a-5p, hsa-miR-582-5p, hsa-miR-1306-5p, hsa-miR-342-3p, hsa-miR-342-15b-3p	Serum EVs	Ultracentrifugation	qRT-PCR	Case-control study (with diagnostic test, GS: NINCDS-ADRDA)
		MCI	3;8	1/2;4/4	76.91±4.4; 78.91±6.3	23±5.66; 30±1.08					
		HC	23;36	10/13;21/14	73.07±7.57; 78.55±6.52	29±1.25; 30±1.08					
Fiandaca et al., 2015	America	AD	57;24	30/27; 12/12	79.5±6.05	22.9±1.02	P-S396-tau, P-T181-tau, $A\beta_{1-42}$	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+NCAM/L1CAM(CD171)	ELISA	Case-control study
		AP	24	12/12	71.8±7.30	-					
		MCI	29	-	-	27.6±0.30					
		HC	57;24	30/27; 12/12	79.6±6.03	-					
Goetzl-2 et al., 2015	America (NIA)	AD	24;16	12/12; 9/7	75.7 ±7.59;83.8±7.54	23.8±0.59;21.3±1.53	LRP6, HSF1, REST	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+L1CAM(CD171)	ELISA	Case-control study
		AP	16	9/7	78.2±6.44	29.0±0.31					
		MCI	16;7	-	-	27.1±0.51;24.0±0.90					

		HC	24;16	12/12;9/7	75.1±7.18;78.3±5.99	29.2±0.36;28.7±0.47					
Goetzl-3 et al., 2015	America (NIA)	AD	26;20	13/13;10/10	75.4±7.85;80.1±6.21	22.5±1.48; 21.6±1.07	LAMP-1, HSP70, cathepsin D, ubiquitinylated proteins	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+L1CAM(CD171)	ELISA	Case-control study
		AP	20	10/10	74.0 6 5.95	29.1±0.26					
		MCI	18;9	-	-	28.0±0.40; 25.8±1.30					
		HC	26;20	13/13;10/10	75.8±7.91;73.8±5.93	-					
Kapogiannis-1 et al., 2015	America (NIA) (JHSF)	AD	26;22	13/13;10/12	74.3±7.48; 78 . 0±1.69	8.81 ±1.00	Total IRS-1, P-serine 312-IRS-1, P-pan-tyrosine-IRS-1, the ratio of P-serine 312-IRS-1 to P-pan-tyrosine-IRS-1 (R)	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+NCAM/L1CAM(CD171)	ELISA	Case-control study
		PC-AD	22	10/12	-	-					
		MCI	16	-	-	28.3±0.48					
		HC	26	13/13	74.3±7.48	-					
Lugli et al., 2015	America	AD	46	-	-	-	miR-185-5p, miR-342-3p, miR-141-3p, miR-548at-5p, miR-342-5p, miR-4772-3p, miR-23b-3p, miR-138-5p, miR-24-3p, miR-24-3p, miR-29b-3p, miR-3916, miR-125b-5p, miR-338-3p, miR-3065-5p, miR-139-5p, miR-152-3p, miR-150-5p, miR-5001-3p, miR-659-5p, miR-3613-3p	Plasma EVs	Ultracentrifugation	qRT-PCR	Case-control study
		HC	41	-	-	-					

Goetzl-4 et al., 2016	America (NIA) (UCSF)	AD	12;9	6/6;7/2	74.4±1.98; 87.8±2.50	26.3±0.99;21.4±1.60	Synaptotagmin, synaptopodin, synaptophysi, GAP-43, neurogranin	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+L1CAM(CD171)	ELISA	Case-control study
		AD_1	9	7/2	81.7± 2.06	28.7±0.47					
		HC	12;9	6/6;7/2	74.4±1.98; 82.2±2.28	29.8±0.11;28.3±0.96					
Goetzl-5 et al., 2016	America (UCSF)	AD	12	-	-	-	BACE-1, γ-secretase, sAPPβ, sAPPα, Septin-8, GDNF	Plasma NDE/ADE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+GLAST(ACSA-1) /L1CAM	ELISA	Case-control study
		HC	10	-	-	-					
Winston-1 et al., 2016	America	AD	10	-	-	-	P-T181-tau, P-S396-tau, $A\beta_{1-42}$, REST, neurogranin	plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+L1CAM(CD171)	ELISA	Case-control study (with diagnostic test, GS: CSF-A β)
		MCI	20	-	68.7±7.76	28.82±0.33					
		ADC	20	-	75.35±6.82	17.67± 0.70					
		HC	10	-	-	-					
Goetzl-1 et al., 2018	America (NIA) (JHSF)	AD	28;18	16/12;8/10	73.1±1.44; 78.2±1.75	25.6±0.83;20.2±1.50	NPTX2, NRXN2a, AMPA4, NLGN1	Plasma NDE/ADE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+L1CAM(CD171)/ GLAST	ELISA	Case-control study
		AD_1	18	8/10	69.4±1.71	28.7±0.47					
		HC	28;18	16/12;8/10	73.2±1.47; 70.1±1.66	29.7±0.13;28.3±0.96					

Goetzl-7 et al., 2018	America (NIA) (JHSF)	AD	28;16	16/12;7/9	73.2±1.47;79.1±1.83	25.6±0.83;21.6±1.46	IL-6, TNF-α, IL-1β, C1q, C4b, C3d, factor B, factor D, factor B fragment Bb, C3b, C5b-C9 TCC, MBL, CD59, CD46, CR1, DAF, factor I	Plasma ADE/NDE	ExoQuick (System Biosciences, Mountain View, CA) +ACSA-1/L1CAM	ELISA	Case-control study
		AD_1	16	7/9	70.3±1.75	28.5±0.62					
		HC	28;16	16/12;7/9	73.2±1.47; 69.7± 2.04	29.7±0.13 ;29.1± 0.87					
Winston-2 et al., 2018	America (NIA)	MCI	61	39/22	70.2±2.3	27.9±0.64	neurogranin, synaptophysin, synaptotagmin, synaptopodin, P-S396-tau, $A\beta_{1-42}$, GAP43	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+L1CAM(CD171)	ELISA	Case-control study (with diagnostic test, GS: NIA-AA diagnostic guideline)
		HC	76	47/29	67.8±2.3	29.1±0.33					
Yang et al., 2018	China	AD	107	66/41	74.15±7.93	-	miR-135a, miR-193b, miR-384	Serums EVs	Total Exosome Isolation kit (Invitrogen Life Technologies)	qRT-PCR	Case-control study (with diagnostic test, GS: NINCDS-ADRDA)
		MCI	101	59/42	61.63±7.32	-					
		HC	228	-	-	-					
Agliard et al., 2019	Italy	AD	24	16/8	77.67±1.40	21.91±0.91	SNAP-25	Serum NDE	ExoQuick® (System Biosciences, LCC, USA)+L1CAM (CD171)	ELISA	Case-control study (with diagnostic test, GS: NINCDS-ADRDA)
		HC	17	13/4	76.47±1.49	28.73±0.43					
Cha et al., 2019	America	AD	31	-	-	-	miR-212, miR-132	Plasma NDE	Ultracentrifugation	qRT-PCR	Case-control study (with diagnostic test, GS: modified NIA-Reagan Diagnosis)
		MCI	16	-	-	-					
		HC	16	-	-	-					
Gámez-Valero et al., 2019	Germany	AD	10	6/4	70.5 (61 – 85)	18.6 (12–23)	hsa-miR-23a-3p, hsa-mir-126-3p, hsa-let-7i-5p,	Plasma EVs	SEC	qRT-PCR	Case-control study (with diagnostic test, GS: NIA-AA

		HC	15	10/5	70.5 (61 – 85)	27.6 (26–30)	hsa-mir-151a-3p				diagnostic guideline)
Goetzl-6 et al., 2019	America (NIA)	AD	24;15	15/9;8/7	73.1±1.60;84.5±1.70	26.1±0.90;24.3±0.90	HGF, FGFs-2 FGFs-13, IGF-1	Plasma NDE/ADE/CSPG4E	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+CSPG4/GLAST/L1CAM	ELISA	Case-control study
		AD_1	15	8/7	80.7±1.84	28.3±0.46					
		HC	24;15	15/9;8/7	73.1±1.82;80.2±1.78	29.3±0.19;29.4±0.62					
Haddadl et al., 2019	America	ES (AD)	17	3/14	79.41±1.3	23.65±0.9	CML	Serum EVs	Total Exosome Isolation reagent (Invitrogen™ by Life Technologies Inc., Carlsbad, CA, USA)	ELISA	Case-control study
		MS (AD)	9	2/7	80.78±1.3	21.11±1.3					
		LS (AD)	12	3/9	81.58±1.6	-					
		MCI	14	10/4	73.29±1.4	27.07±0.52					
		HC	13	11/2	69.68±1.15	29.62±0.2					
Jia-1 et al., 2019	China	AD	28;73	16/12;42/32	20.3± 3.2;19.6± 3.1	20.3±3.2;21±8.14	$A\beta_{1-42}$, T-tau, P-T181-tau	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+NCAM	ELISA	Case-control study (with diagnostic test, GS: NIA-AA diagnostic guideline)
		aMCI	25;71	13/12;39/32	26.5±0.4;26.2±0.4	26.5±0.4;29±2.03					
		HC	29;72	15/14;37/35	29.2±1.2 ;29.3±1.2	29.2±1.2 ;30±1.08					
Kapogiannis-2 et al., 2019	America (BLSA) (ADRC)	AD	128;35	68/60;18/17	79.09±7.02;74.03 ±8.73	-	P-serine 312-IRS-1, P-pan-tyrosine-IRS-1	Plasma EVs	ExoQuick exosome precipitation solution (EXOQ; System	MSD	Case-control study (with diagnostic test, GS:

		HC	218;29	110/108;23/6	76.2±7.36;72.14±7.86	-			Biosciences, CA)+L1CAM		NINCDS-ADRDA)
Winston-3 et al., 2019	America (JHSF)	AD	20	8/12	71.1±6.90	23.9 ± 0.75	C1q, C4b, factor D, factor B fragment Bb, C5b, C3b, C5b-C9 TCC, MBL, CD46, CD59, CR1, DAF	plasma ADE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+GLAST/ACSA-1	ELISA	Case-control study
		MCIS	20	7/13	68.7±7.76	29.0±0.26 ;28.8±0.33					
		MCIC	20	9/11	75.4±6.82	27.4±0.29;17.7±0.70					
		HC	20	8/12	70.8±5.34	29.2±0.45					
Aharon et al., 2020	Israel	Mild AD	20	7/13	68.2±7.06	25.47±2.74	IFN-γ, CCL5, CXCL1, IL-2/8, AgRP, VEGF-D, VEGFR-2/3; ANG-1; PDGF-BB; TPO, Let-7g-5p, miR-126-3p, miR-142-3p, miR-146a-5p, miR-223-3p	Plasma EVs	Ultracentrifugation	Protein array, qRT-PCR	Case-control study
		Moderate AD	13	4/9	69.69±10.57	17.31±2.93					
		Severe AD	9	5/4	62.3±9.87	10.89±2.15					
		HC	19	11/8	65.84±6.53	-					
Ellegaard et al., 2020	Denmark	AD	10	6/4	72 (68–74)	25.5 (21.3–27.3)	Neurology and Inflammation panel TGF-α | CCL20, CLEC1B | CCL11	Plasma EVs	Ultracentrifugation	PEA	Case-control study (with diagnostic test, GS: ICD_{10}, NINCDS-ADRDA, Petersen criteria)
		MCI	10	8/2	72 (69–76)	27.5 (26.0–30.0)					
		HC	10	6/4	65 (65–66)	-					
Gu et al., 2020	China	AD	31	23/8	68.58±8.04	15.93±6.61	$A\beta_{1-42}$, P-T181-tau, P-S396-tau, IL-6, MMP-9	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System	ELISA	Case-Control study (with diagnostic test, GS: IWG-2)

		HC	15	10/5	64.80±6.00	27.67±1.72			Biosciences, CA)+L1CAM		
Jia-2 et al., 2020	China	AP	160	90/70	60	29.2 (1.2)	GAP-43, neurogranin, SNAP-25, synaptotagmin 1	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+NCAM	ELISA	Case-control study (with diagnostic test, GS: NIA-AA diagnostic guideline)
		HC	160	86/74	60	29.1±1.2					
		Non-mutation	59	32/27	48.5 (9.7)	29.9±1.0					
		Mutation-carriers	62	F 30 (50.8)	48.9 (9.5)	29.9±0.8					
Li et al., 2020	China	AD	13;18	7/6;10/8	74.2 (6.1) ;68.2 (8.7)	20.0 (1.9); 19.3 (2.5)	hsa-miR-1306-5p, hsa-miR-342-3p, hsa-15b-3p, hsa-miR-361-5p, hsa-miR-30e-5p, hsa-miR-93-5p, hsa-miR-15a-5p, hsa-miR-143-3p, hsa-miR-335-5p, hsa-miR-106b-5p, hsa-miR-101-3p, hsa-miR-424-5p, hsa-miR-106a-5p, hsa-miR-18b-5p, hsa-miR-3065-5p, hsa-miR-20a-5p, hsa-miR-582-5p, and hsa-miR-451, $A\beta_{1-42}$, T-tau, P-T181-tau, P-S396-tau	Serum EVs	Total Exosome Isolation Reagent kit (ThermoFisher Scientific, Carlsbad, CA, USA)	qRT-PCR	Case-control study (with diagnostic test, GS: NINCDS-ADRDA)
		MCI	10;18	6/4;12/6	76.6 (5.4) ;69.6 (10.3)	10.8 (3.9) ;12.6 (4.7)					
		HC	10;18	6/4;12/6	74.6 (5.0) ;69.1 (10.6)	27.7 (1.9); 27.7 (1.9)					
Nam et al., 2020	South Korea	Mild AD	18	-	-	-	$A\beta_{1-42}$, T-tau, P-T181-tau, P-S202-tau	Serum NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+ CD171/L1CAM	ELISA	Case-control study (with diagnostic test, GS: DSM-IV criteria)
		MCI	29	-	-	-					

		HC	23	-	-	-					
Nie et al., 2020	China	AD	5	4/1	67.8	13.8	miR-423-5p, miR-369-5p, miR-23a-3p, miR-204-5p, miR-125a-5p, miR-1468-5p, miR-375, let-7e-5p	Plasma NDE	ExoQuick Plasma prep and Exosome isolation kit (SBI, Palo Alto, United States)	Sequencing	Case-control study
		HC	34	14/20	33.2	-					
Perrotte et al., 2020	Canada	Mild AD	12	11/1	75.6±1.3	24.00±0.52	T-tau, APP, $A\beta_{1-42}$, P-T181-tau	Plasma NDE	Total Exosome Isolation kit (Invitrogen Life Technologies))+L1CAM	Luminex	Case-control study (with diagnostic test, GS: NINCDS-ADRDA)
		Moderate AD	12	8/4	79,1±1.1	19.90±1.39					
		Severe AD	12	10/2	83.00±1.60	-					
		MCI	12	11/1	75.33±1.19	27.90±0.31					
		HC	12	9/3	68.8±1.5	29.42±0.29					
Serpente et al., 2020	Italy	AD	40	25/15	72.8(2.8)	-	miR-23a-3p, miR-223-3p, miR-100-3p, miR-190-5p	plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+CD171/L1CAM/CD81	BSA, qRT-PCR	Case-control study
		HC	40	18/22	66.6 (5.1)	-					
Sun et al., 2020	China	AD	60	29/31	72.1±10.1	18.2±6.5	P-S396-tau, $A\beta_{1-42}$	plasma exosomes	ExoQuick Exosome Precipitation (ExoQuickTM SBI)	ELISA	Case-control study (with diagnostic test, GS:

		HC	60	28/32	69.6±6.3	27.1±1.3					NINCDS-ADRDA)
Wang et al., 2020	China	AD	72	38/34	73.7±7.7	13.7±6.7	BACE1-AS, BACE1 mRNA, BC200, 51A	Plasma exosomes	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, LLC)	qRT- PCR	Case-control study
		HC	62	39/23	71.8±8.1	27.2±2.1					
Zhang et al., 2020	China	AD	24	17/7	67.8±8.2	16.3±6.1	TDP-43	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+L1CAM(CD171)	ELISA	Case-control study
		HC	15	10/5	64.8±6.0	27.7±1.7					
Zhao et al., 2020	China	AD	88	50/38	67.7 (4.2)	17.0 (2.1)	$A\beta_{1-42}$	Plasma NDE	ExoQuick exosome precipitation solution (EXOQ; System Biosciences, CA)+L1CAM(CD171)	ELISA	Case-control study (with diagnostic test, GS: NIA-AA diagnostic guideline)
		MCI	87	47/40	66.2 (4.3)	25.7 (1.4)					
		MCI-c	16	9/7	68.5 (3.8)	19.3 (1.7)					
		MCI-nc	62	34/28	68.3 (4.1)	25.8 (1.4)					
		HC	80	44/36	67.3 (4.7)	29.3 (0.7)					

In some articles, it may include two parts of research, such as discovery set and validation set. Some studies also include two tests on two different groups of people. In addition, some studies also include cross-sectional and longitudinal observation studies. Therefore, ";" aims to distinguish the basic characteristics of two groups of different subjects. The sign after the author means that there are several authors with the same last name or some studies with the same first author, and the number represents the order of the article, which aims to distinguish these studies. "-": Not described. AD: Alzheimer's disease; ADC: patients who transitioned within 36 months from MCI to AD; ADE: astrocyte-derived exosomes; ADRC: Alzheimer Disease Research Center; AD_1: subjects with cognitively intact before the diagnosis of Alzheimer disease; AgRP: agouti-related peptide; AIBL: Australian Imaging: Biomarkers and Lifestyle; aMCI: amnestic mild cognitive impairment; AMPA4: GluA4-containing glutamate; ANG-1: angiopoietin 1; ANM: AddNeuroMed; AP: preclinical Alzheimer disease; Aβ: β-amyloid; BACE-1: β-site amyloid precursor protein-cleaving enzyme 1; BACE1-AS: β-site amyloid precursor protein cleaving enzyme-1-antisense transcript; BC200: brain cytoplasmic 200 RNA; BLSA: Baltimore Longitudinal Study of Aging; BSA: Bulked Segregant Analysis; CCL5: chemokine (C-C motif) ligand 5; CLEC1B: c-type lectin domain family 1 member B; CML: N-(1-carboxymethyl)-L-lysine; CR1: complement receptor 1; CSF: cerebrospinal fluid; CSPG4E: exosomes derived from chondroitin sulfate proteoglycan 4 type neural precursor cells; CXCL1: C-X-C motif chemokine ligand 1; C1q: complement 1q; DAF: decay-accelerating factor; DSM-IV: Diagnostic and Statistical Manual of Mental Disorders; ELISA: enzyme linked immunosorbent assay; ES(AD): early-stage of Alzheimer's disease; EVs: extracellular vesicles; F: Female; FGFs-2: fibroblast growth factors-2; GAP-43: growth-associated protein 43; GDNF: glial-derived neurotrophic factor; GLAST: glutamine aspartate

transporter; GS: gold standard; HC: healthy control subjects; HGF: hepatocyte growth factor; HSF1: the heat-shock factor-1; HSP70: heat-shock protein 70; ICD_{10}: International Classification of Diseases and Related Health Problems 10th Edition; IFN-γ: interferon-gamma; IGF-1: type 1 insulin-like growth factor; IL-2: interleukin 2; IRS-1: insulin receptor substrate 1; IWG-2: International Working Group-2; JHSF: Jewish Home of San Francisco; LAMP-1: lysosome-associated membrane protein 1; LRP6: low-density lipoprotein receptor-related protein 6; LS(AD): late-stage of Alzheimer's disease; L1CAM: L1 cell adhesion molecule; M: male; MBL: mannose-binding lectin; MCI: mild cognitive impairment; MCIC: subjects with mild cognitive impairment on entry into the study that converted to dementia by the end of the 3-year study; MCI-c: MCI converters; MCI-nc: MCI non-converters; MCIS: subjects with mild cognitive impairment that was stable over the 3-year study; MMP-9: matrix metalloproteinase-9; MMSE: Mini-Mental State Examination; miR: microRNA; MS(AD): moderate-stage of Alzheimer's disease; MSD: Mesoscale Discovery; N: number; NCAM: neural cell adhesion molecule; NDE: neurally-derived exosomes; NIA-AA: National Institute on Aging and Alzheimer's Association; NINCDS-ADRDA: National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association; NLGN1: neuroligin 1; NPTX2: neuronal pentraxin 2; NRXN2a: neurexin 2α; PC-AD: subjects with cognitively intact before the diagnosis of Alzheimer disease; PDGF-BB: platelet-derived growth factor BB; PEA: Proximity Extension Assay; P-tau: phosphorylated tau; qRT-PCR: Real-time Quantitative polymerase chain reaction; REST: repressor element 1-silencing transcription; sAPP: soluble amyloid precursor protein; SEC: Size Exclusion Chromatography; SNAP-25: synaptosomal-associated protein-25; TCC: terminal complement complex; TDP-43:TAR DNA binding protein of 43 kDa; TGF-α: transforming growth factor-α; TNF-α: tumor necrosis factor-α; TPO: thrombopoietin; T-tau: total tau; UCSF: University of California San Francisco; VEGFR-2: vascular endothelial growth factor receptor 2.